

Project 7

MongoDB & Mongo Express on Kubernetes — Deployments, Services, ConfigMaps & Secrets

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Aim

To deploy MongoDB and Mongo Express on a local Kubernetes (Minikube) cluster, using Deployments, Services, ConfigMaps, and Secrets to manage the database credentials and networking securely.

PHASE 1 Create the Secret and ConfigMap

1 Encode Your Credentials (The Industry Way)

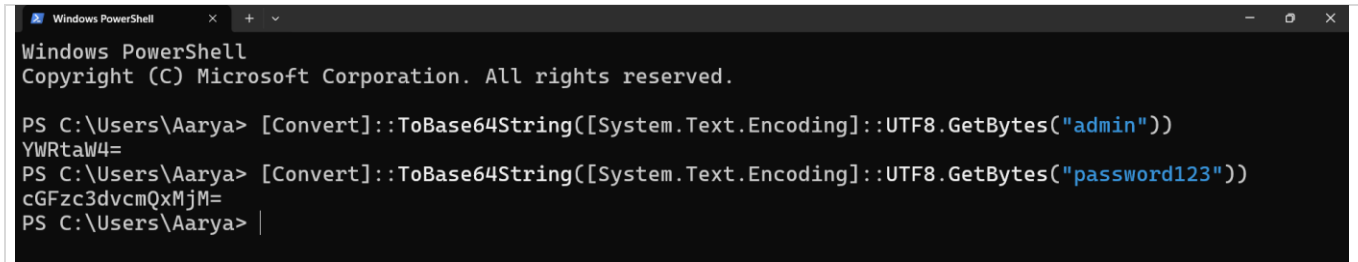
Kubernetes Secrets require your passwords to be encoded in base64. Since you are using PowerShell, you can generate these securely right in your terminal.

Run these two commands to encode the username admin and the password password123:

POWERSHELL

```
[Convert]::ToBase64String([System.Text.Encoding]::UTF8.GetBytes("admin"))  
[Convert]::ToBase64String([System.Text.Encoding]::UTF8.GetBytes("password123"))
```

Copy the generated gibberish strings — you will need them in a second!



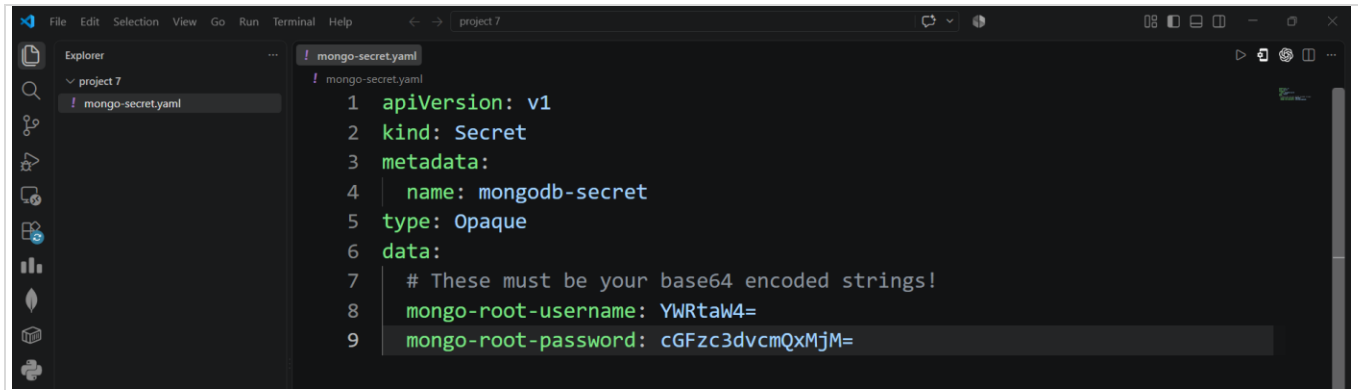
```
Windows PowerShell  
Copyright (C) Microsoft Corporation. All rights reserved.  
  
PS C:\Users\Aarya> [Convert]::ToBase64String([System.Text.Encoding]::UTF8.GetBytes("admin"))  
YWRTaW4=  
PS C:\Users\Aarya> [Convert]::ToBase64String([System.Text.Encoding]::UTF8.GetBytes("password123"))  
cGFzc3dvcmQxMjM=  
PS C:\Users\Aarya> |
```

2 Create the Secret File

Create a new file named mongo-secret.yaml and paste this configuration. Replace the <ENCODED_USER> and <ENCODED_PASSWORD> with the base64 strings you just generated:

YAML — mongo-secret.yaml

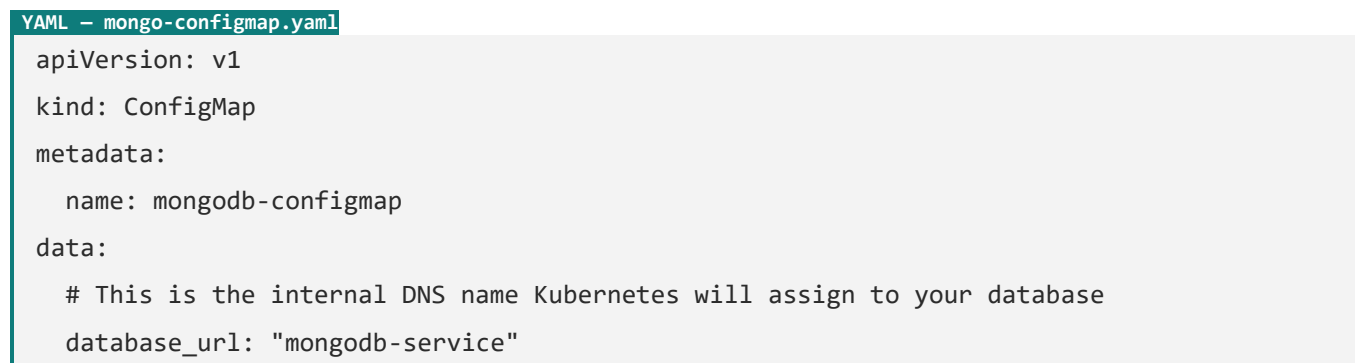
```
apiVersion: v1  
kind: Secret  
metadata:  
  name: mongodb-secret  
type: Opaque  
data:  
  # These must be your base64 encoded strings!  
  mongo-root-username: <ENCODED_USER>  
  mongo-root-password: <ENCODED_PASSWORD>
```

A screenshot of the Visual Studio Code editor interface. The Explorer sidebar on the left shows a project named 'project 7' containing a file named 'mongo-secret.yaml'. The main editor window displays the content of 'mongo-secret.yaml' with the following YAML structure:

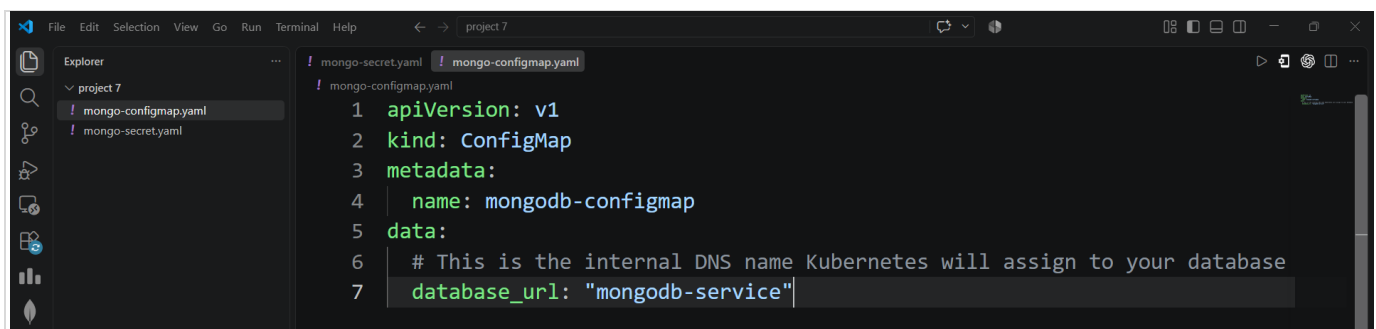
```
1 apiVersion: v1
2 kind: Secret
3 metadata:
4   name: mongodb-secret
5 type: Opaque
6 data:
7   # These must be your base64 encoded strings!
8   mongo-root-username: YWRtaW4=
9   mongo-root-password: cGFzc3dvcmQxMjM=
```

3 Create the ConfigMap File

Create another file named mongo-configmap.yaml. We will use this later to tell Mongo Express exactly where to find the MongoDB database on the internal network.

A screenshot of the Visual Studio Code editor interface showing the 'mongo-configmap.yaml' file. The Explorer sidebar on the left shows 'project 7' with two files: 'mongo-configmap.yaml' and 'mongo-secret.yaml'. The main editor window displays the content of 'mongo-configmap.yaml' with the following YAML structure:

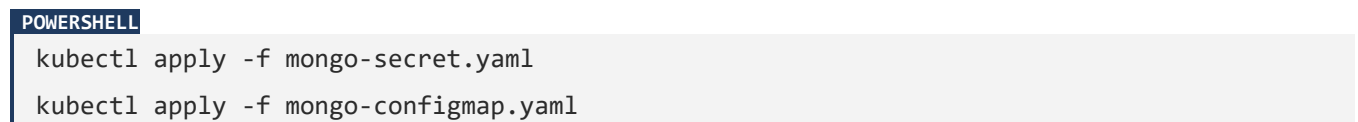
```
YAML - mongo-configmap.yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: mongodb-configmap
data:
  # This is the internal DNS name Kubernetes will assign to your database
  database_url: "mongodb-service"
```

A screenshot of the Visual Studio Code editor interface showing the 'mongo-configmap.yaml' file. The Explorer sidebar on the left shows 'project 7' with two files: 'mongo-configmap.yaml' and 'mongo-secret.yaml'. The main editor window displays the content of 'mongo-configmap.yaml' with the following YAML structure:

```
1 apiVersion: v1
2 kind: ConfigMap
3 metadata:
4   name: mongodb-configmap
5 data:
6   # This is the internal DNS name Kubernetes will assign to your database
7   database_url: "mongodb-service"
```

4 Apply the Configurations

Run these commands in PowerShell to inject your configurations into the cluster:

A screenshot of a PowerShell terminal window. It contains two commands to apply the Kubernetes configurations:

```
POWERSHELL
kubectl apply -f mongo-secret.yaml
kubectl apply -f mongo-configmap.yaml
```

PHASE 2 Start the Minikube Cluster

1 Start Docker Desktop

1. Open your Windows Start Menu and click on **Docker Desktop**.
2. Wait a minute or two. Look at the bottom-left corner of the Docker Desktop window and wait until it shows a green icon and says **Engine running**.

Docker must be fully awake before Minikube can use it.

2 Start Minikube

Open a fresh PowerShell terminal and run this command to boot up your Kubernetes cluster inside Docker:

POWERSHELL

```
minikube start --driver=docker
```

Wait for this process to completely finish and return you to the prompt.

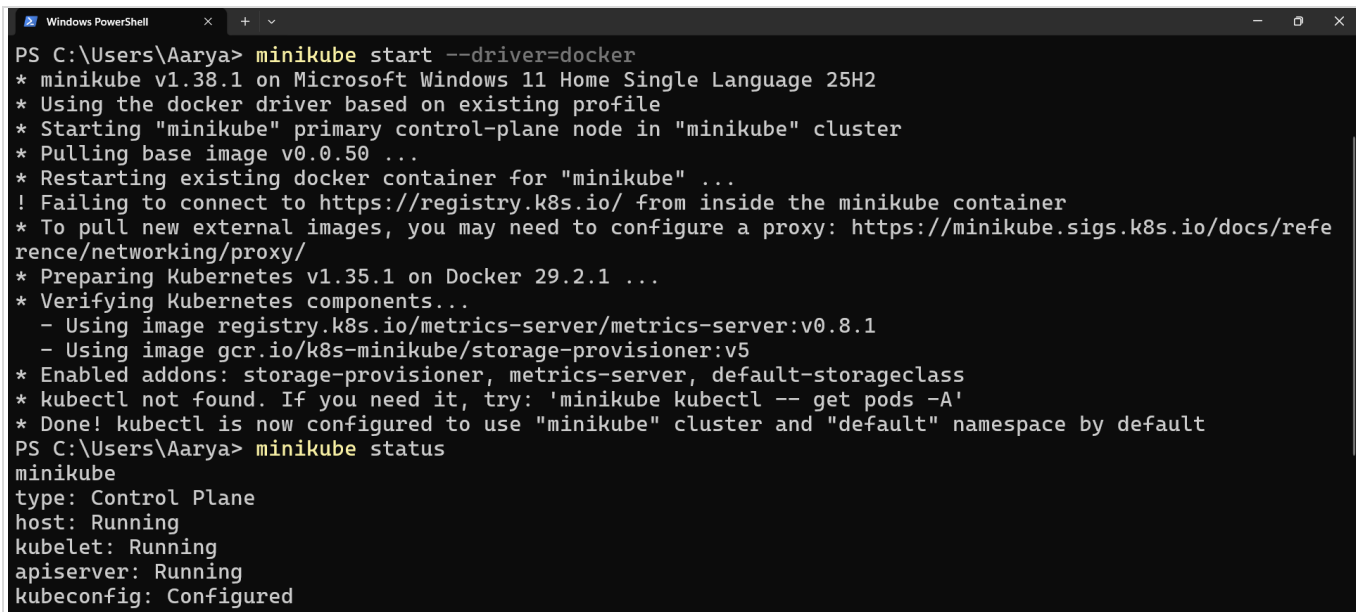
3 Verify the Cluster

Make sure the cluster is healthy and ready to accept commands:

POWERSHELL

```
minikube status
```

You should see statuses saying Running and Configured.



```
PS C:\Users\Aarya> minikube start --driver=docker
* minikube v1.38.1 on Microsoft Windows 11 Home Single Language 25H2
* Using the docker driver based on existing profile
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Restarting existing docker container for "minikube" ...
! Failing to connect to https://registry.k8s.io/ from inside the minikube container
* To pull new external images, you may need to configure a proxy: https://minikube.sigs.k8s.io/docs/reference/networking/proxy/
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Verifying Kubernetes components...
  - Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
  - Using image gcr.io/k8s-minikube/storage-provisioner:v5
* Enabled addons: storage-provisioner, metrics-server, default-storageclass
* kubectl not found. If you need it, try: 'minikube kubectl -- get pods -A'
* Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\Aarya> minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
```

4 Navigate to Your Folder

Before you run any configurations, use the `cd` command in PowerShell to navigate to the exact folder where you want to save your project files. For example:

POWERSHELL

```
cd F:\SEM_7\my-git-practice\Day_6\
```

Replace that path with wherever you want to store this MongoDB project.

5 Apply the Configurations

Now that the engine is running and you are in the correct folder, you can create the files we discussed and apply them.

If you haven't created the files yet, make sure you create mongo-secret.yaml and mongo-configmap.yaml first, then run:

POWERSHELL

```
kubectl apply -f mongo-secret.yaml
kubectl apply -f mongo-configmap.yaml
```

Once you do that, you can create the mongo-database.yaml file from my previous message and run:

POWERSHELL

```
kubectl apply -f mongo-database.yaml
```

```
PS C:\Users\Aarya> cd 'F:\SEM_7\Devops_Lab\project 7\'
PS F:\SEM_7\Devops_Lab\project 7> kubectl apply -f mongo-secret.yaml
secret/mongodb-secret created
PS F:\SEM_7\Devops_Lab\project 7> kubectl apply -f mongo-configmap.yaml
configmap/mongodb-configmap created
PS F:\SEM_7\Devops_Lab\project 7> |
```

PHASE 3 Deploy MongoDB

1 Create the Database Configuration File

In the same folder where your secrets are, create a new file named mongo-database.yaml and paste this complete architecture block:

YAML – mongo-database.yaml

```
# 1. The Virtual Storage
apiVersion: v1
kind: PersistentVolumeClaim
metadata:
  name: mongodb-pvc
spec:
  accessModes:
    - ReadWriteOnce
  resources:
    requests:
      storage: 1Gi
---
# 2. The Database Container
```

```

apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongodb-deployment
  labels:
    app: mongodb
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mongodb
  template:
    metadata:
      labels:
        app: mongodb
    spec:
      containers:
        - name: mongodb
          image: mongo:latest
          ports:
            - containerPort: 27017
          # Pulling credentials securely from your Secret
          env:
            - name: MONGO_INITDB_ROOT_USERNAME
              valueFrom:
                secretKeyRef:
                  name: mongodb-secret
                  key: mongo-root-username
            - name: MONGO_INITDB_ROOT_PASSWORD
              valueFrom:
                secretKeyRef:
                  name: mongodb-secret
                  key: mongo-root-password
          # Setting enterprise resource limits
          resources:
            limits:
              cpu: "500m"
              memory: "512Mi"

```

```

    requests:
      cpu: "200m"
      memory: "256Mi"
    # Connecting the virtual storage
    volumeMounts:
      - name: mongodb-data
        mountPath: /data/db
    volumes:
      - name: mongodb-data
        persistentVolumeClaim:
          claimName: mongodb-pvc
---
# 3. The Internal Network Service
apiVersion: v1
kind: Service
metadata:
  name: mongodb-service # This name MUST match your ConfigMap!
spec:
  selector:
    app: mongodb
  ports:
    - protocol: TCP
      port: 27017
      targetPort: 27017

```

File Edit Selection View Go Run Terminal Help project 7

Explorer

- project 7
 - ! mongo-configmap.yaml
 - ! mongo-database.yaml
 - ! mongo-secret.yaml

! mongo-database.yaml

```
! mongo-database.yaml
1 # 1. The Virtual Storage
2 apiVersion: v1
3 kind: PersistentVolumeClaim
4 metadata:
5   name: mongodb-pvc
6 spec:
7   accessModes:
8     - ReadWriteOnce
9   resources:
10     requests:
11       storage: 1Gi
12 ---
13 # 2. The Database Container
14 apiVersion: apps/v1
15 kind: Deployment
16 metadata:
17   name: mongodb-deployment
18   labels:
19     app: mongodb
20 spec:
21   replicas: 1
22   selector:
23     matchLabels:
24       app: mongodb
25   template:
26     metadata:
27       labels:
28         app: mongodb
29     spec:
30       containers:
31         - name: mongodb
32           image: mongo:latest
33           ports:
34             - containerPort: 27017
```

```
! mongo-database.yaml
! mongo-database.yaml
20 spec:
25   template:
29     spec:
31       - name: mongodb
34         containerPort: 27017
35         # Pulling credentials securely from your Secret
36         env:
37           - name: MONGO_INITDB_ROOT_USERNAME
38             valueFrom:
39               secretKeyRef:
40                 name: mongodb-secret
41                 key: mongo-root-username
42           - name: MONGO_INITDB_ROOT_PASSWORD
43             valueFrom:
44               secretKeyRef:
45                 name: mongodb-secret
46                 key: mongo-root-password
47         # Setting enterprise resource limits
48         resources:
49           limits:
50             cpu: "500m"
51             memory: "512Mi"
52           requests:
53             cpu: "200m"
54             memory: "256Mi"
55         # Connecting the virtual storage
56         volumeMounts:
57           - name: mongodb-data
58             mountPath: /data/db
59         volumes:
60           - name: mongodb-data
61             persistentVolumeClaim:
62               claimName: mongodb-pvc
63         ---
```

```
! mongo-database.yaml
! mongo-database.yaml
20 spec:
25   template:
29     spec:
60       - name: mongodb-data
61         persistentVolumeClaim:
63         ---
64     # 3. The Internal Network Service
65     apiVersion: v1
66     kind: Service
67     metadata:
68       name: mongodb-service # This name MUST match your ConfigMap!
69     spec:
70       selector:
71         app: mongodb
72       ports:
73         - protocol: TCP
74           port: 27017
75           targetPort: 27017
```


2 Apply the Configuration

Run this command in PowerShell to build the storage, the database, and the network service all at once:

POWERSHELL

```
kubectl apply -f mongo-database.yaml
```

```
PS F:\SEM_7\Devops_Lab\project 7> kubectl apply -f mongo-database.yaml
persistentvolumeclaim/mongodb-pvc created
deployment.apps/mongodb-deployment created
service/mongodb-service created
PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1     Running   4 (6m41s ago)  2d2h
backend-deployment-66cb79749-bjc8k  1/1     Running   5 (6m41s ago)  2d2h
frontend-deployment-74b9f7fc98-slvvs 1/1     Running   2 (6m3s ago)   2d2h
mongodb-deployment-594d6f6c44-6w7fg  0/1     ContainerCreating 0              11s
PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1     Running   4 (7m4s ago)  2d2h
backend-deployment-66cb79749-bjc8k  1/1     Running   5 (7m4s ago)  2d2h
frontend-deployment-74b9f7fc98-slvvs 1/1     Running   2 (6m26s ago) 2d2h
mongodb-deployment-594d6f6c44-6w7fg  0/1     ContainerCreating 0              34s
PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1     Running   4 (7m51s ago) 2d2h
backend-deployment-66cb79749-bjc8k  1/1     Running   5 (7m51s ago) 2d2h
frontend-deployment-74b9f7fc98-slvvs 1/1     Running   2 (7m13s ago) 2d2h
mongodb-deployment-594d6f6c44-6w7fg  0/1     ContainerCreating 0              81s
PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1     Running   4 (32m ago)   2d3h
backend-deployment-66cb79749-bjc8k  1/1     Running   5 (32m ago)   2d3h
frontend-deployment-74b9f7fc98-slvvs 1/1     Running   2 (32m ago)   2d3h
mongodb-deployment-594d6f6c44-6w7fg  1/1     Running   1 (23m ago)   26m
PS F:\SEM_7\Devops_Lab\project 7> |
```

3 Verify the Build

Run this command to check on the status of your database pod:

POWERSHELL

```
kubectl get pods
```

```
PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1     Running   4 (6m41s ago)  2d2h
backend-deployment-66cb79749-bjc8k  1/1     Running   5 (6m41s ago)  2d2h
frontend-deployment-74b9f7fc98-slvvs 1/1     Running   2 (6m3s ago)   2d2h
mongodb-deployment-594d6f6c44-6w7fg  0/1     ContainerCreating 0              11s
PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1     Running   4 (7m4s ago)  2d2h
backend-deployment-66cb79749-bjc8k  1/1     Running   5 (7m4s ago)  2d2h
frontend-deployment-74b9f7fc98-slvvs 1/1     Running   2 (6m26s ago) 2d2h
mongodb-deployment-594d6f6c44-6w7fg  0/1     ContainerCreating 0              34s
PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1     Running   4 (7m51s ago) 2d2h
backend-deployment-66cb79749-bjc8k  1/1     Running   5 (7m51s ago) 2d2h
frontend-deployment-74b9f7fc98-slvvs 1/1     Running   2 (7m13s ago) 2d2h
mongodb-deployment-594d6f6c44-6w7fg  0/1     ContainerCreating 0              81s
PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1     Running   4 (32m ago)   2d3h
backend-deployment-66cb79749-bjc8k  1/1     Running   5 (32m ago)   2d3h
frontend-deployment-74b9f7fc98-slvvs 1/1     Running   2 (32m ago)   2d3h
mongodb-deployment-594d6f6c44-6w7fg  1/1     Running   1 (23m ago)   26m
PS F:\SEM_7\Devops_Lab\project 7> |
```

PHASE 4 Deploy Mongo Express

1 Create the Mongo Express Configuration

Create a new file named mongo-express.yaml in your folder and paste this architecture block:

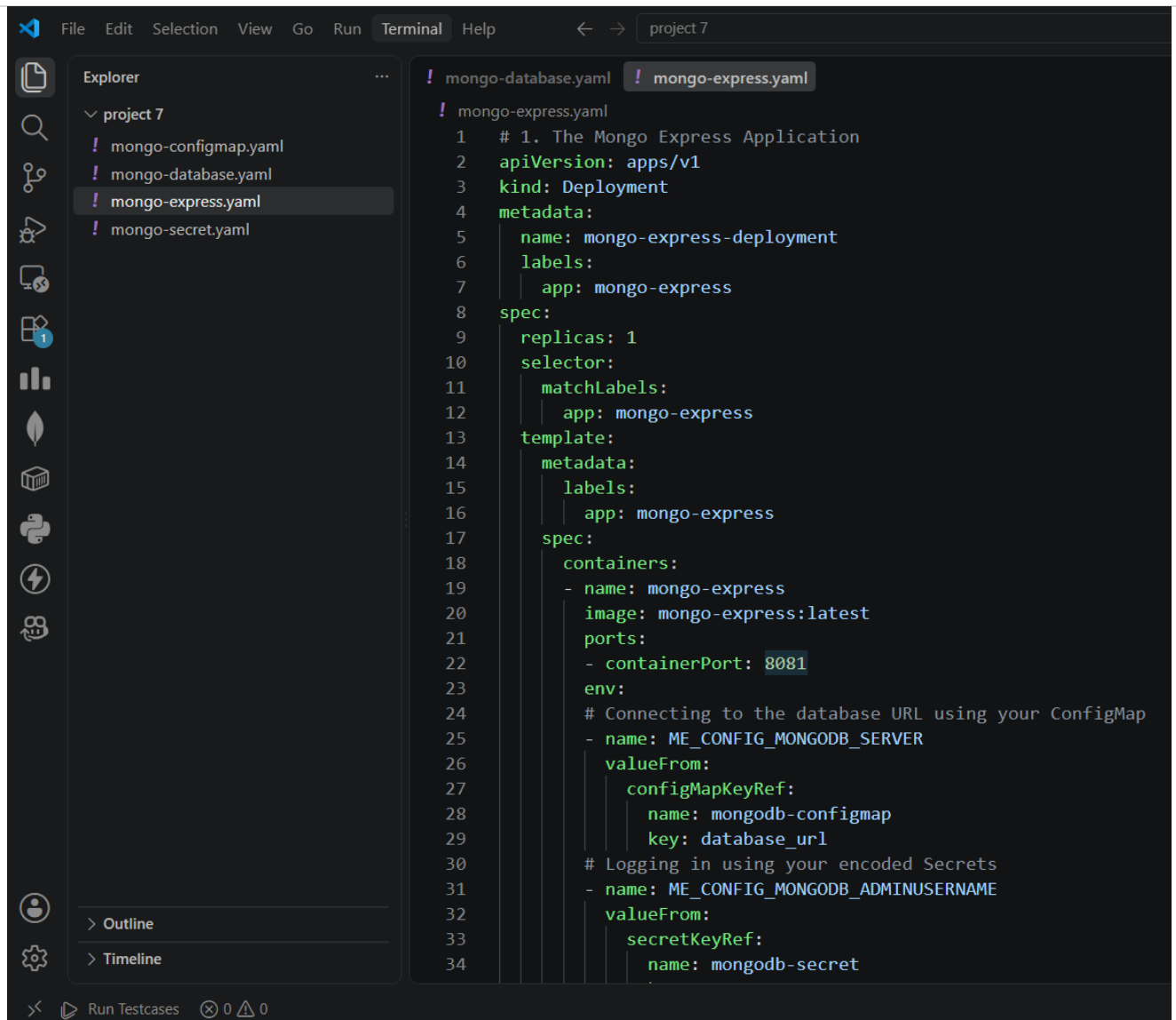
YAML — mongo-express.yaml

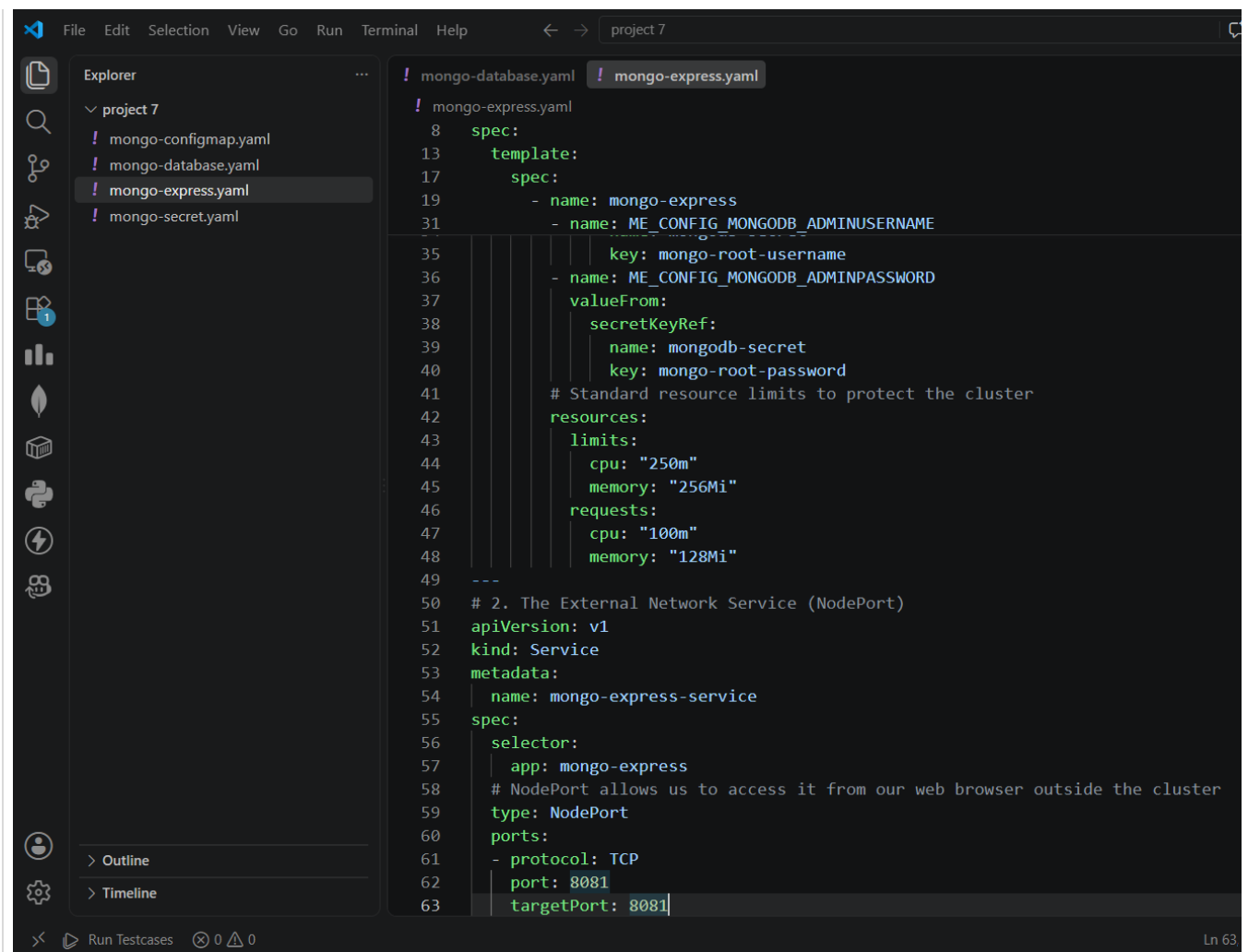
```
# 1. The Mongo Express Application
apiVersion: apps/v1
kind: Deployment
metadata:
  name: mongo-express-deployment
  labels:
    app: mongo-express
spec:
  replicas: 1
  selector:
    matchLabels:
      app: mongo-express
  template:
    metadata:
      labels:
        app: mongo-express
    spec:
      containers:
        - name: mongo-express
          image: mongo-express:latest
          ports:
            - containerPort: 8081
          env:
            # Connecting to the database URL using your ConfigMap
            - name: ME_CONFIG_MONGODB_SERVER
              valueFrom:
                configMapKeyRef:
                  name: mongodb-configmap
                  key: database_url
            # Logging in using your encoded Secrets
            - name: ME_CONFIG_MONGODB_ADMINUSERNAME
              valueFrom:
                secretKeyRef:
```

```

        name: mongodb-secret
        key: mongo-root-username
    - name: ME_CONFIG_MONGODB_ADMINPASSWORD
      valueFrom:
        secretKeyRef:
          name: mongodb-secret
          key: mongo-root-password
# Standard resource limits to protect the cluster
resources:
  limits:
    cpu: "250m"
    memory: "256Mi"
  requests:
    cpu: "100m"
    memory: "128Mi"
---
# 2. The External Network Service (NodePort)
apiVersion: v1
kind: Service
metadata:
  name: mongo-express-service
spec:
  selector:
    app: mongo-express
# NodePort allows us to access it from our web browser outside the cluster
type: NodePort
ports:
  - protocol: TCP
    port: 8081
    targetPort: 8081

```





2 Apply the Configuration

Run this command in PowerShell to deploy the UI and expose it:

POWERSHELL

```
kubectl apply -f mongo-express.yaml
```

3 Verify Both Pods Are Running

Let's make sure both the database and the UI are successfully running side-by-side:

POWERSHELL

```
kubectl get pods
```

You should see both mongodb-deployment-... and mongo-express-deployment-... showing a status of Running. It might take a minute for Mongo Express to pull the image.

```
Windows PowerShell
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured

PS F:\SEM_7\Devops_Lab\project 7> kubectl apply -f mongo-express.yaml
error: the path "mongo-express.yaml" does not exist
PS F:\SEM_7\Devops_Lab\project 7> kubectl apply -f mongo-express.yaml
deployment.apps/mongo-express-deployment created
service/mongo-express-service created
PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY    STATUS              RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1      Running             4 (77m ago) 2d4h
backend-deployment-66cb79749-bjc8k  1/1      Running             5 (77m ago) 2d4h
frontend-deployment-74b9f7fc98-slvvs 1/1      Running             2 (76m ago) 2d4h
mongo-express-deployment-645f678fc7-tw6s8 0/1      ContainerCreating   0           6s
mongodb-deployment-594d6f6c44-6w7fg  1/1      Running             1 (67m ago) 70m

PS F:\SEM_7\Devops_Lab\project 7> kubectl get pods
NAME                                READY    STATUS              RESTARTS   AGE
backend-deployment-66cb79749-26x7f  1/1      Running             4 (78m ago) 2d4h
backend-deployment-66cb79749-bjc8k  1/1      Running             5 (78m ago) 2d4h
frontend-deployment-74b9f7fc98-slvvs 1/1      Running             2 (78m ago) 2d4h
mongo-express-deployment-645f678fc7-tw6s8 1/1      Running             0           97s
mongodb-deployment-594d6f6c44-6w7fg  1/1      Running             1 (69m ago) 72m

PS F:\SEM_7\Devops_Lab\project 7> |
```

4 Open the Dashboard in Your Browser!

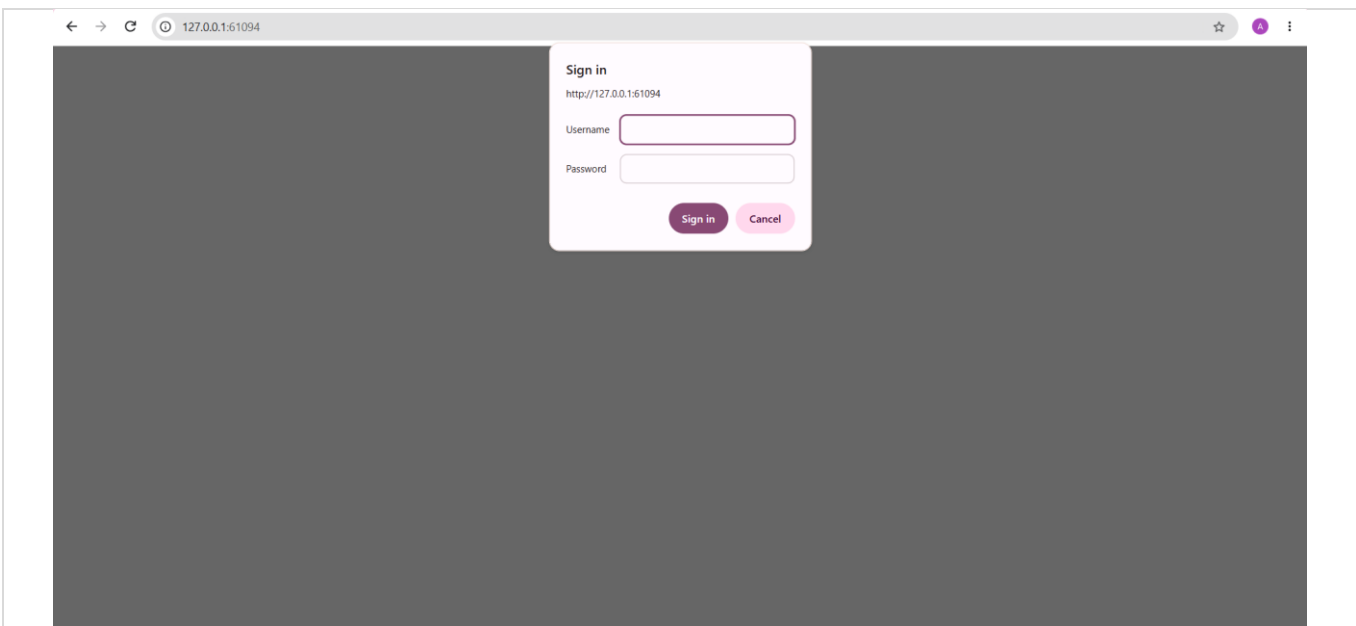
Because we are running this on a local Minikube cluster, we use a special Minikube command to automatically create a secure tunnel from your local network directly to the NodePort service.

Run this command:

POWERSHELL

```
minikube service mongo-express-service
```

Minikube will automatically open your default web browser and drop you right into the Mongo Express dashboard.



127.0.0.1:61094

Mongo Express Database ▾

Mongo Express

Databases

 View	admin	 Del
 View	config	 Del
 View	local	 Del

Server Status

Hostname	mongodb-deployment-594d6f6c44-6w7fg	MongoDB Version	8.2.12
Uptime	4842 seconds	Node Version	18.20.3
Server Time	Tue, 18 Aug 2026 14:44:39 GMT	V8 Version	10.2.154.26-node.37
Current Connections	3	Available Connections	6550
Active Clients	0	Queued Operations	0
Clients Reading	0	Clients Writing	0

127.0.0.1:61094

Mongo Express Database ▾

Server Status

Hostname	mongodb-deployment-594d6f6c44-6w7fg	MongoDB Version	8.2.12
Uptime	4889 seconds	Node Version	18.20.3
Server Time	Tue, 18 Aug 2026 14:45:26 GMT	V8 Version	10.2.154.26-node.37
Current Connections	3	Available Connections	6550
Active Clients	0	Queued Operations	0
Clients Reading	0	Clients Writing	0
Read Lock Queue	0	Write Lock Queue	0
Disk Flushes		Last Flush	
Time Spent Flushing	ms	Average Flush Time	ms
Total Inserts	0	Total Queries	17
Total Updates	1	Total Deletes	0